

# ASyMphony

Finn Cooper – sd22404

# Introduction

## Problem

Assembly is low-level, state changes are hard to see; visual simulations are used.

Existing solutions are predominantly visual or analytical, such as LMC.

## Idea

Playsemble gamifies assembly programming to great success; shows that meaningful consequences enhance engagement.

Live coding gives immediate feedback and inherently exposes state of the system.

Can live coding be applied to assembly programming to enhance learning?

## Objectives

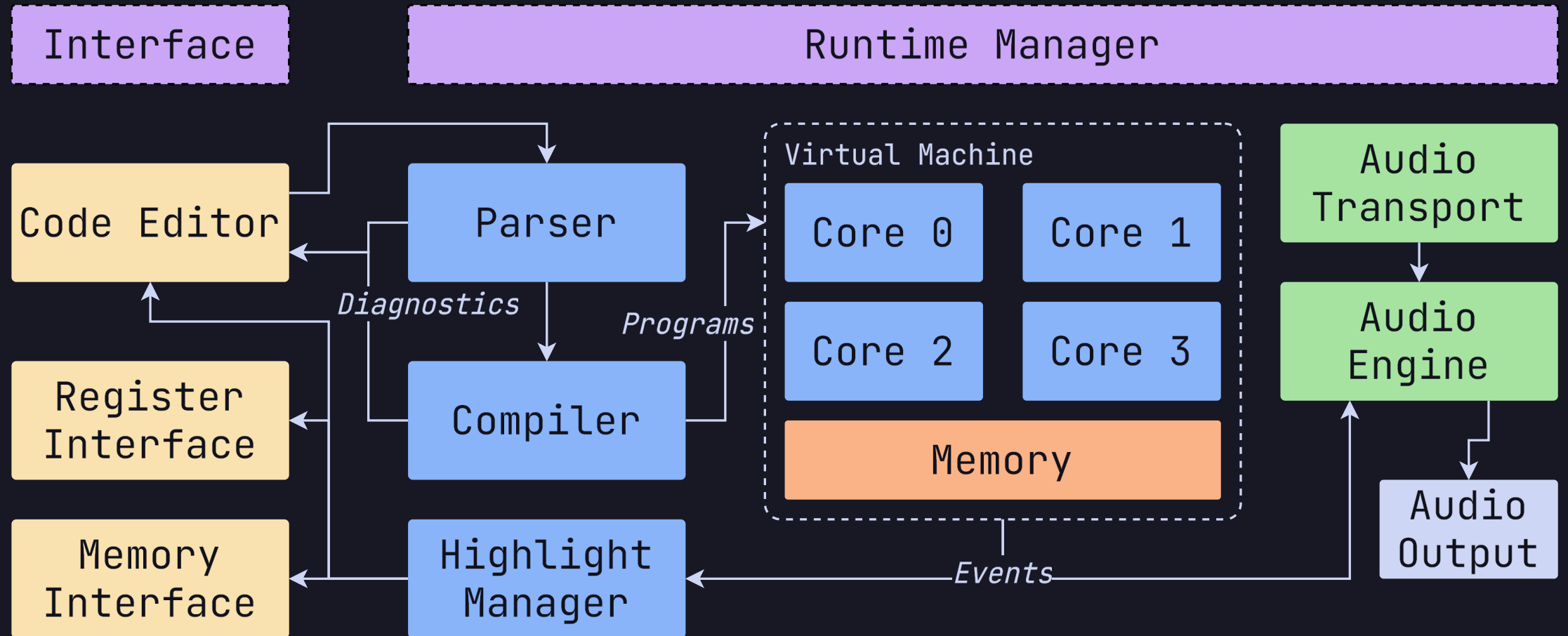
- 01: Accessibility
- 02: Conceptual Transparency
- 03: Musical Engagement
- 04: Musical Expressiveness
- 05: Educational Viability

## Plan

Develop in two stages: a minimum viable product, and a final version.

Build for web to increase accessibility and reduce installation friction.

# Implementation



# Evaluation and Conclusion

## Methods

Jakob Nielsen's Heuristics

Think Aloud Sessions

SUS Questionnaire

Likert Scale Questions

## Results

MVP Nielsen: severe usability issues

MVP SUS: 63.54 – less than “good”

MVP Feedback: solid idea, areas unclear

Final Nielsen: minimal usability issues

Final SUS: 81.04 (+17.50) – “good”

Final Feedback: majorly positive response

## Conclusion

Shows viability as an engaging and usable introduction to assembly programming.

Limited evidence for concrete educational viability and musical expressiveness.

## Limitations

Heuristic analysis used single evaluators

Didn't Evaluate:

- objective improvement in understanding
- musical expressiveness and capabilities
- against other live coding systems
- against other assembly simulators

## Future Work

- assess objective learning improvement
- assess complexity of created music
- add more QoL features to the system